

False-Positive-Constrained IoT Intrusion Detection

A reproducible pilot study using a fixed CICIoT2023 shard

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Study aim

This study asks whether an operating point selected to limit false alarms on validation data retains that behavior on unseen traffic. The focus is not a new classifier; it is the reliability of a threshold chosen under an operational constraint.

Research question

Can validation-based threshold selection retain high attack recall under a 1% false-positive-rate constraint, and does the chosen point continue to meet that constraint on a held-out partition?

Main finding

The rule selected Random Forest at threshold 0.59853. On the held-out partition, attack recall was 98.167% and FPR was 1.208%. The conditional 95% FPR interval was 0.846% to 1.601%. The validation point constraint did not remain satisfied on the held-out data.

Data and methods

CICIoT2023 documents 105 IoT devices and 33 attacks in seven categories. This analysis uses Merged01.csv (712,311 rows; 39 predictors), fixed by SHA-256 8b43d6552a8cafd3b0ca2cedf6464ca3fe644d7fc9bb5dfe906368f1542792fe. Raw data are not redistributed.

After binary label conversion, conflicting feature groups were removed and one representative of each remaining feature vector was retained. The resulting 518,091 observations were split 60/20/20 into development, validation, and held-out partitions using seed 20260920. Imputation and scaling were fitted on development data only.

Logistic Regression and Random Forest were trained with class weighting. For each model, the validation threshold with the highest attack recall under $FPR \leq 1\%$ was retained. If recalls differed by no more than 0.1 percentage points, the smaller fitted model was preferred. The selected configuration was then applied once to the held-out partition.

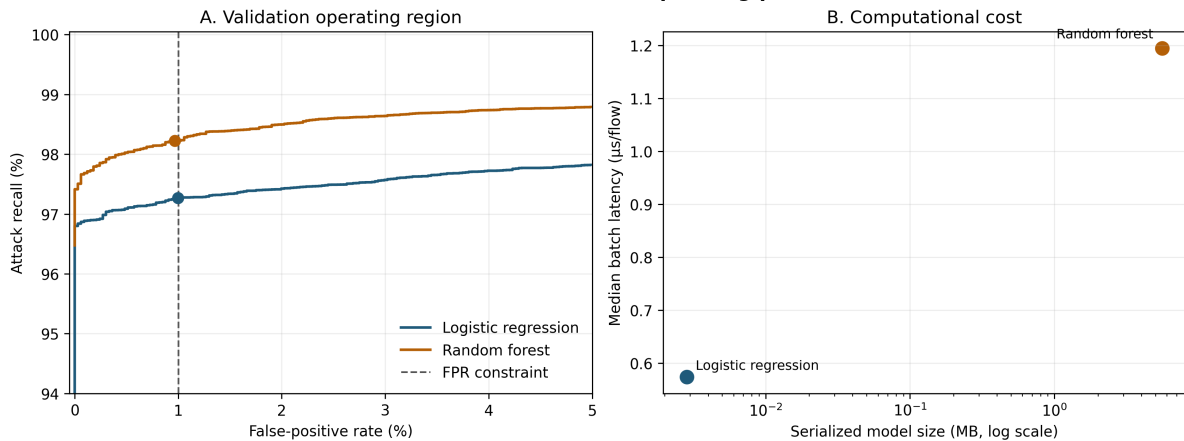
Data audit

Audit item	Observed	Decision
Exact duplicate rows	167,368 (23.50%)	Prevent identical rows from crossing partitions
Duplicate feature rows	194,204 (27.26%)	Treat feature vectors as the sampling unit
Conflicting binary-label groups	16	Exclude from the primary analysis
Attack prevalence	97.673%	Report class-sensitive metrics
Missing / infinite cells	22 / 14	Fit imputation on development data only

Baseline comparison

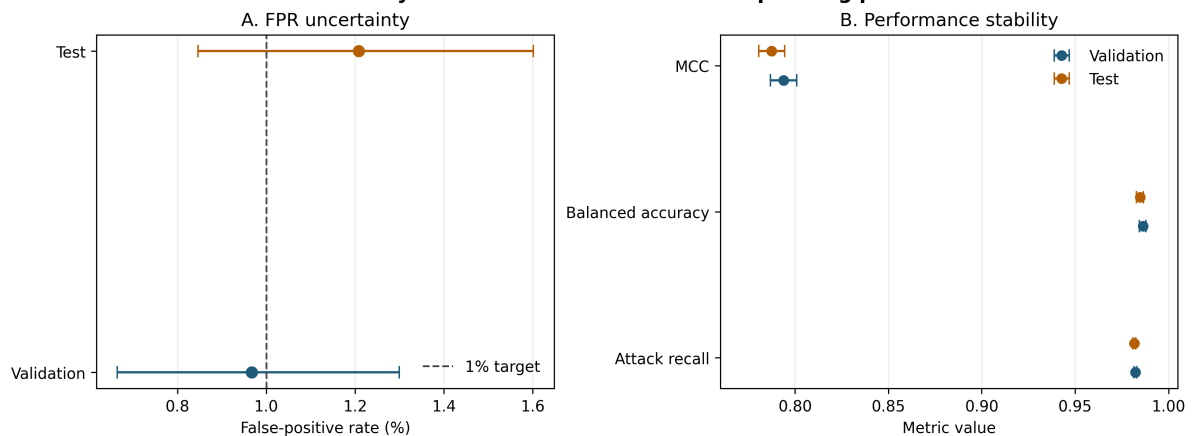
Held-out metric	Logistic Regression	Random Forest
Attack recall	96.880%	98.391%
False-positive rate	0.030%	1.843%
MCC	0.705	0.803
Macro average precision	0.885	0.946
Serialized size	2.9 KB	5.56 MB
Median batch latency	0.574 μ s/flow	1.195 μ s/flow

Validation-constrained model and operating-point selection



At threshold 0.59853, Random Forest validation recall was 98.228%. On held-out data, false positives fell from 61 to 40 relative to threshold 0.50, while missed attacks rose from 1,614 to 1,839.

Uncertainty at the frozen Random Forest operating point



Held-out metric	Observed	Conditional 95% interval
False-positive rate	1.2085%	0.8459% to 1.6012%
Attack recall	98.1667%	98.0839% to 98.2484%
Balanced accuracy	98.4791%	98.2851% to 98.6620%
MCC	0.7875	0.7806 to 0.7945

Limitations. The source shard was globally shuffled and cannot represent a future deployment period. Results come from one split and one training seed. The binary task merges heterogeneous attacks. Timing values are run-specific because hardware metadata were not retained.

Next study. Use device- or time-separated evaluation, repeated training, an external dataset, and an upper confidence bound on FPR. This would test threshold adaptation under changing traffic and compute constraints.

Code and results: <https://github.com/wisam-makki-salim/trustworthy-iot-ids-pilot>

References: [1] Neto, E. C. P. et al. (2023). CICIOT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment. Sensors, 23(13), 5941. <https://doi.org/10.3390/s23135941>. [2] Canadian Institute for Cybersecurity, University of New Brunswick. CIC IoT Dataset 2023. <https://www.unb.ca/cic/datasets/iotdataset-2023.html>